



PROJECT: Data Analysis with Excel Pivot Tables

DATA ANALYST: O. ABIGAEL AJIBUWA

PROJECT 1: U.S VOTERS –2012

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CASE STUDY: (2012 population and voters registration data from the U.S Census Bureau, by state and age group)

- ▶ Concepts: Calculated Fields; Value Setting (% of Column, % of Parent)
- ▶ Dimensions: State, Age
- ▶ Measures: Total Population, Citizen Population, Registered Voters, Confirmed Voters

1. How many states had a Voter Population % below 55%? Which states?
2. How many confirmed voters in California were over 65 years old in 2012? What percentage does that represent out of the total confirmed voters in California? What percentage out of the confirmed voters in the entire country?
3. Show both Citizen Population and Confirmed Voters by Age, as % of Column Total. What percentage of the citizen population do 45- to 64-year-olds represent? What percentage of the confirmed voter population?
4. Create a new calculated field named "Voter Turnout" (Confirmed Voters/Registered Voters), formatted as a percentage with 2 decimal points. Which state had the highest voter turnout rate? What about among 18-24-year-old voters specifically?
5. As a politician seeking to improve voter turnout rates among young adults (18-24), which states would you target first?

PROJECT 1: U.S VOTERS –2012

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1. How many states had a Voter Population % below 55%? Which states ?

Age	(All)				
Row Labels	Sum of Citizen Population2	Sum of Confirmed Voters2	Sum of Confirmed Voters	Sum of Voter Population %	
Texas	16,062,000	69.88%	8,643,000	53.81%	
Arkansas	2,110,000	9.09%	1,124,000	53.27%	
Oklahoma	2,734,000	11.57%	1,431,000	52.34%	
Hawaii	930,000	3.89%	481,000	51.72%	
West Virginia	1,443,000	5.57%	689,000	47.75%	
Grand Total	23,279,000	100.00%	12,368,000	53.13%	

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2. How many confirmed voters in California were over 65 years old in 2012? What percentage does that represent out of the total confirmed voters in California? What percentage out of the confirmed voters in the entire country?

 California	13,463,000	100.00%	10.13%
18 to 24	1,447,000	10.75%	1.09%
25 to 34	2,070,000	15.38%	1.56%
35 to 44	2,118,000	15.73%	1.59%
45 to 64	4,926,000	36.59%	3.71%
65+	2,902,000	21.56%	2.18%

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3. Show both Citizen Population and Confirmed Voters by Age, as % of Column Total. What percentage of the citizen population do 45- to 64-year-olds represent? What percentage of the confirmed voter population?

State	(All)				
Row Labels	Citizen Population	Sum of Citizen Population	Confirmed Voters	Sum of Confirmed Voters	
18 to 24	27,537,000	12.80%	11,351,000	8.54%	
25 to 34	35,474,000	16.49%	18,978,000	14.27%	
35 to 44	34,266,000	15.93%	20,965,000	15.77%	
45 to 64	76,641,000	35.63%	52,013,000	39.12%	
65+	41,167,000	19.14%	29,645,000	22.30%	
Grand Total	215,085,000	100.00%	132,952,000	100.00%	

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4. Create a new calculated field named "Voter Turnout" (Confirmed Voters/Registered Voters), formatted as a percentage with 2 decimal points. Which state had the highest voter turnout rate? What about among 18–24-year-old voters specifically?

Age	(All)					
Row Labels		Citizen Population	Sum of Citizen Population	Confirmed Voters	Sum of Confirmed Voters	Sum of Voter Turnout
Colorado		3,543,000	1.65%	2,495,000	1.88%	94.69%
Wisconsin		4,247,000	1.97%	3,128,000	2.35%	94.27%
Minnesota		3,903,000	1.81%	2,859,000	2.15%	92.67%
Wyoming		419,000	0.19%	247,000	0.19%	92.16%
Delaware		643,000	0.30%	420,000	0.33%	91.60%

Row Labels		Citizen Population	Sum of Citizen Population	Confirmed Voters	Sum of Confirmed Voters	Sum of Voter Turnout
18 to 24		27,537,000	12.80%	11,351,000	8.54%	76.87%
Wisconsin		513,000	0.24%	287,000	0.22%	93.18%
Colorado		461,000	0.21%	259,000	0.19%	93.17%
Minnesota		510,000	0.24%	303,000	0.23%	92.10%
South Carolina		451,000	0.21%	243,000	0.18%	91.70%

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5. As a politician seeking to improve voter turnout rates among young adults (18-24), which states would you target first?

Row Labels	 Citizen Population	Sum of Citizen Population	Confirmed Voters	Sum of Confirmed Voters	Sum of Voter Turnout
 18 to 24	27,537,000	100.00%	11,351,000	100.00%	76.87%
Tennessee	590,000	2.14%	207,000	1.82%	67.65%
Missouri	520,000	1.89%	223,000	1.96%	67.17%
Illinois	1,122,000	4.07%	396,000	3.49%	65.13%
Alaska	61,000	0.22%	20,000	0.18%	64.52%
Montana	79,000	0.29%	33,000	0.29%	62.26%
South Dakota	94,000	0.34%	31,000	0.27%	59.62%
Texas	2,249,000	8.17%	572,000	5.04%	58.91%
Arkansas	281,000	1.02%	70,000	0.62%	55.56%
West Virginia	162,000	0.59%	37,000	0.33%	47.44%
Grand Total	27,537,000	100.00%	11,351,000	100.00%	76.87%

PROJECT 2: San Francisco Salaries

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CASE STUDY 2: (Salary Information from San Francisco government employees, 2011-2013)

- ▶ Concepts: Sorting & Filtering; Grouping; Calculated Fields; Table layouts
- ▶ Dimensions: Employee name, Year, Employee ID, Job Title
- ▶ Measures: Base pay, Overtime pay, Other pay

1. Filter and sort the Pivot to show the 5 employees who earned the highest Base Pay in 2011. Who were they?
2. Add a calculated field named "% Other Pay" (Other Pay/Total Pay), formatted as a percentage with 1 decimal. How many job titles earned only Other pay in 2012?
3. Among employees with $\geq \$100k$ Base Pay in 2012, Did any employee earn more than 50% of their salary from Other Pay? If so, who?
4. Clear all filters and pull in Job Category and Job Title as row labels (Titles sorted alphabetically), then group any titles including the word "Curator" into a new category called "Curator". How many employees held some sort of Curator position in either 2012 or 2013? Among those, who earned the highest average base pay?

PROJECT 2: San Francisco Salaries

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1. Filter and sort the Pivot to show the 5 employees who earned the highest Base Pay in 2011. Who were they?

Year	2011	
Row Labels		Sum of Base Pay
John J Loftus		\$274,126.50
Naomi M Kelly		\$270,641.50
Barbara A Garcia		\$270,591.04
George Gascon		\$252,221.06
Kenneth A Lombardi		\$239,247.00

PROJECT 2: San Francisco Salaries

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2. Add a calculated field named "% Other Pay" (Other Pay/Total Pay), formatted as a percentage with 1 decimal. How many job titles earned only Other pay in 2012?

Year	2012
Row Labels	Sum of % Other pay
Auditor Iii	100.0%
Assistant Recreation Director	100.0%
Asst Materials Coordinator	100.0%
Ct Comp Facilities Coord	100.0%
Auto Svc Wrk Asst Sprv	100.0%
Industrial Injury Investigator	100.0%
Representative, Human Rights Commission	100.0%
Senior Storekeeper	100.0%
Recreation Director	99.8%
Senior Community Development Specialist Ii	77.2%

PROJECT 2: San Francisco Salaries

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3. Among employees with $\geq \$100k$ Base Pay in 2012, Did any employee earn more than 50% of their salary from Other Pay? If so, who?

Year	2012		
Row Labels		Sum of Base Pay	Sum of Other Pay
Anna L Cuthbertson		\$116,478.01	\$124,034.01
Frank E Lee		\$115,178.00	\$110,794.56
Garrett Dowd		\$114,116.13	\$91,314.02
James Bosch		\$110,661.20	\$111,446.20
Edward Browne		\$112,401.08	\$50,391.05
Glenn Ortega		\$112,401.07	\$49,607.84

PROJECT 2: San Francisco Salaries

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4. Clear all filters and pull in Job Category and Job Title as row labels (Titles sorted alphabetically), then group any titles including the word "Curator" into a new category called "Curator". How many employees held some sort of Curator position in either 2012 or 2013? Among those, who earned the highest average base pay?

Year	(Multiple Items)	
Job Category	Employee Name	Average of Base Pay
Curator	Dennis G Sharp	\$81,065.54
	Sonya Knudsen	\$75,891.36
	Allison Cummings	\$75,790.04
	Ramekon O'Arwisters	\$74,076.00
	Robert D Harris	\$72,816.03
	Dennis Sharp	\$72,362.53
	Ann H.A. Carroll	\$62,900.51
	Aimee Le Duc	\$54,149.92
	Allison Harding	\$25,247.40

CASE STUDY 2: (Shark attack records from 1900 – 2016)

- ▶ Concepts: Date Grouping, Value Settings (% of Column, Running Total), PivotCharts
 - ▶ Dimensions: Case Number, Date, Type, Country, Area, Location, Activity, Name, Gender, Age, Injury, Fatal(Y/N), Species, Investigator or Source
 - ▶ Measures: None
1. Show the count of attacks by country -- which 3 countries had the highest number of reported attacks over the past 5 years (2012-2016)? During this period, what % of reported attacks occurred in Spain?
 2. Drag the "Area" field to the PivotTable row labels, change the Report Layout to Outline, and filter to show the top 5 areas by count of Case Number, by country. Where in South Africa were shark attacks most frequently reported over these 5 years?
 3. Replace "Area" with "Type" and show the Count of Case Number values as % of Parent Total for each country. What % of attacks in New Zealand were unprovoked? How many cases?
 4. Filter the Pivot for the USA only and create a PivotChart to show the count of shark attacks by "Area". Use a column chart, and hide all field buttons. Try changing the chart type to a Donut or Bar Chart instead.
 5. Remove the Year filter to show records from the full sample. What was Darren Good doing when he was attacked? When did this happen? (Note: I did not edit this record).

PROJECT 3: Shark Attack records

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1. Show the count of attacks by country -- which 3 countries had the highest number of reported attacks over the past 5 years (2012-2016)? During this period, what % of reported attacks occurred in Spain?

Date	(All)	▼	
Years	(Multiple Items)	▼	
Months	(All)	▼	
Row Labels	▼↓	Count of Case Number	Count of Case Number2
USA		313	51.74%
Australia		128	21.16%
South Africa		30	4.96%
Bahamas		19	3.14%
Reunion		12	1.98%
French Polynesia		10	1.65%
Spain		9	1.49%
New Caledonia		8	1.32%

PROJECT 3: Shark Attack records

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2. Drag the "Area" field to the PivotTable row labels, change the Report Layout to Outline, and filter to show the top 5 areas by count of Case Number, by country. Where in South Africa were shark attacks most frequently reported over these 5 years?

Date	(All)		
Years	(Multiple Items)		
Months	(All)		
Country	Area	Count of Case Number	Count of Case Number2
[-] South Africa		30	4.96%
	Western Cape Province	13	2.15%
	Eastern Cape Province	12	1.98%
	KwaZulu-Natal	3	0.50%
	Unknown	2	0.33%
[-] Bahamas		19	3.14%
	Unknown	8	1.32%

PROJECT 3: Shark Attack records

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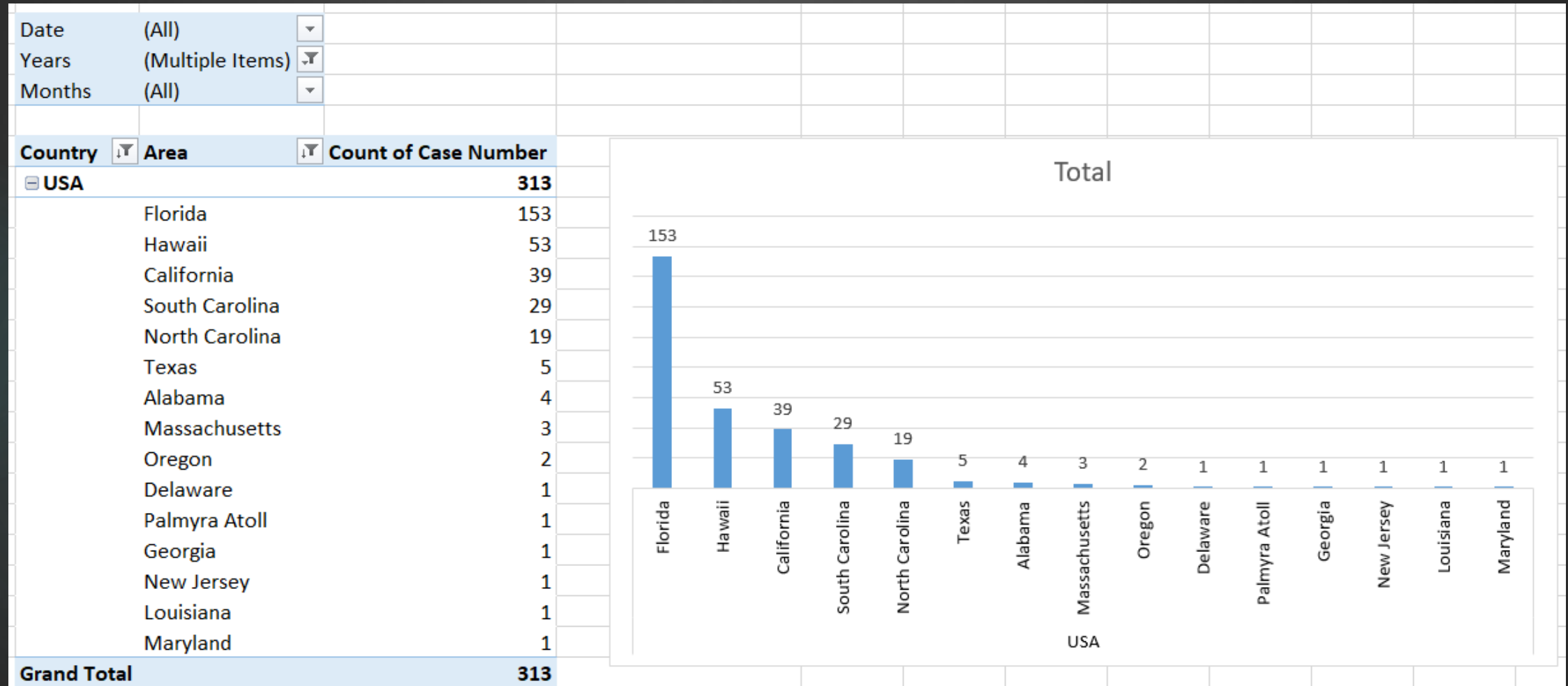
3. Replace "Area" with "Type" and show the Count of Case Number values as % of Parent Total for each country. What % of attacks in New Zealand were unprovoked? How many cases?

Date	(All)	▼		
Years	(Multiple Items)	▼		
Months	(All)	▼		
Country	▼	Type	▼	Count of Case Number
				Count of Case Number2
[-] New Zealand				7
	Provoked			1
	Unprovoked			6
[-] Mexico				5
	Provoked			1
	Unprovoked			4

PROJECT 3: Shark Attack records

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4. Filter the Pivot for the USA only and create a PivotChart to show the count of shark attacks by "Area". Use a column chart and hide all field buttons. Try changing the chart type to a Donut or Bar Chart instead.



PROJECT 3: Shark Attack records

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5. Remove the Year filter to show records from the full sample. What was Darren Good doing when he was attacked? When did this happen? (Note: I did not edit this record).

Name	Activity	Date
Darren Good	Dived naked into the water on a bet	26/02/1996
Grand Total		

PROJECT 4: Stock Market Data

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CASE STUDY: (3-month sample of stock market data for 500 publicly-traded companies)

- ▶ Concepts: Sorting & Filtering; Grouping; conditional Formatting (Highlight Cells, data bars); Value Setting (% Difference from)
- ▶ Dimensions: Date, Symbol
- ▶ Measures: Open, High, Low, Close, volume

1. Create a calculated field named "Daily Spread" (High - Low), formatted as currency with two decimal places. On which date in the sample did Amazon (AMZN) see the largest price spread? (note: you may have to remove the Daily Change field)
2. Sort dates oldest to newest and conditionally format the Daily Spread field as a Color Scale, from white (lower values) to green (higher values). Since a large price spread isn't necessarily good or bad, edit the color scale to show shades from white to blue (instead of green).
3. Drag in another instance of Volume and show values as Difference From the previous date. When did Google (GOOG) see the largest day-over-day gain in trading volume? The largest drop?

PROJECT 4: Stock Market Data

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1. Create a calculated field named "Daily Spread" (High - Low), formatted as currency with two decimal places. On which date in the sample did Amazon (AMZN) see the largest price spread? (note: you may have to remove the Daily Change field)

Symbol	AMZN		
Row Labels	Sum of High	Sum of Low	Sum of Daily Spread
23/10/2009	\$119,65	\$110,62	\$9,03
26/10/2009	\$125,68	\$118,49	\$7,19
16/09/2009	\$90,98	\$85,90	\$5,08
27/10/2009	\$124,26	\$119,42	\$4,84
05/11/2009	\$120,05	\$116,25	\$4,79

PROJECT 4: Stock Market Data

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- Sort dates oldest to newest, and conditionally format the Daily Spread field as a Color Scale, from white (lower values) to green (higher values). Since a large price spread isn't necessarily good or bad, edit the color scale to show shades from white to blue (instead of green).

Symbol	AMZN			
Row Labels	Sum of High	Sum of Low	Sum of Daily Spread	
21/08/2009	\$85,06	\$83,54	\$1,52	
24/08/2009	\$85,59	\$84,24	\$1,35	
25/08/2009	\$86,34	\$83,95	\$2,39	
26/08/2009	\$84,76	\$83,38	\$1,38	
27/08/2009	\$84,85	\$83,14	\$1,71	
28/08/2009	\$84,99	\$82,57	\$2,42	
31/08/2009	\$81,95	\$80,35	\$1,60	
01/09/2009	\$82,42	\$79,00	\$3,42	
02/09/2009	\$80,15	\$77,80	\$2,35	
03/09/2009	\$78,96	\$77,51	\$1,45	
04/09/2009	\$79,78	\$77,63	\$2,15	
09/09/2009	\$82,63	\$80,50	\$2,13	
10/09/2009	\$84,07	\$82,30	\$1,77	
11/09/2009	\$84,90	\$83,76	\$1,14	
14/09/2009	\$84,57	\$83,46	\$1,11	
15/09/2009	\$84,41	\$82,79	\$1,62	
16/09/2009	\$90,98	\$85,90	\$5,08	
17/09/2009	\$91,19	\$89,00	\$2,19	

PROJECT 4: Stock Market Data

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3. Drag in another instance of Volume and show values as Difference From the previous date. When did Google (GOOG) see the largest day-over-day gain in trading volume? The largest drop?

Symbol	GOOG			
Row Labels	Sum of High	Sum of Low	Sum of Daily Spread	Sum of Volume
06/10/2009	\$499.37	\$491.70	\$7.67	6,065
07/10/2009	\$518.99	\$497.81	\$21.18	21,447
08/10/2009	\$523.25	\$513.34	\$9.91	-5,711
09/10/2009	\$521.51	\$514.50	\$7.01	-15,667
12/10/2009	\$525.76	\$519.32	\$6.44	5,845
13/10/2009	\$527.46	\$521.38	\$6.08	-2,821
14/10/2009	\$535.58	\$530.00	\$5.58	2,228
15/10/2009	\$536.90	\$527.27	\$9.63	28,353
16/10/2009	\$554.75	\$544.53	\$10.22	27,454
19/10/2009	\$553.60	\$548.73	\$4.87	-56,256
20/10/2009	\$552.95	\$540.70	\$12.25	8,243
21/10/2009	\$559.35	\$549.00	\$10.35	-3,713

PROJECT 5: Baseball Team Stats

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CASE STUDY: (Major Leagues Baseball team statistics by season, 1995 – 2015)

- ▶ Concepts: Sorting & Filtering; PivotCharts and Slicers; Calculated Fields
- ▶ Dimensions: Season start, Year, League, Team (Short), Team (Full), Division, DivWin, WCWin, LggWin, WSWin
- ▶ Measures: G, W, L, RS, AB, H, 2B, 3B, HR, BB, SO, SB, CS, RA, ER, ERA, CG, SHO, SV

1. Create a view showing RS and RA by Team, then create calculated fields named "Net Runs" ($RS - RA$) and "HR per Game" (HR/G). Which team had the highest Net Run total over the entire sample? What about just the 2015 season?
2. Update the Pivot to show data for the Red Sox by year. Which years did they win the Division (DivWin)? The Wild Card (WCWin)? The World Series (WSWin)?
3. Show home runs (HR) by year, for the entire sample (all teams), and drag in a second instance as the % Difference From the previous year. In which season did overall home run totals decrease the most Y-o-Y?
4. Update your layout to show Net Runs and HR per Game as Values, Year as Row Labels, and Team (Full) as a Slicer. Select the Baltimore Orioles from the slicer, then insert a Pivot Chart formatted as a Combo Chart, with Net Runs plotted as columns on the primary axis, and HR per Game plotted as a line on the secondary axis. Do the two metrics follow a similar trend?

PROJECT 5: Baseball Team Stats

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1. Create a view showing RS and RA by Team, then create calculated fields named "Net Runs" (RS - RA) and "HR per Game" (HR/G). Which team had the highest Net Run total over the entire sample? What about just the 2015 season?

Row Labels	RS	RA	Net Runs	HR	G	HR per Game
New York Yankees	17,787	15,162	2,625	4183	3383	1.2364765
Atlanta Braves	15,661	13,711	1,950	3607	3383	1.06621342
Boston Red Sox	17,591	15,783	1,808	3935	3383	1.163168785
St. Louis Cardinals	15,981	14,545	1,436	3496	3382	1.033707865
Los Angeles Dodgers	14,847	13,914	933	3234	3383	0.955956252
Oakland Athletics	16,144	15,255	889	3658	3382	1.081608516
Cleveland Indians	16,811	16,005	806	3804	3381	1.125110914

Year	2015						
Row Labels	RS	RA	Net Runs	HR	G	HR per Game	
Toronto Blue Jays	891	670	221	232	162	1.432098765	
St. Louis Cardinals	647	525	122	137	162	0.845679012	
Houston Astros	729	618	111	230	162	1.419753086	
Pittsburgh Pirates	697	596	101	140	162	0.864197531	

PROJECT 5: Baseball Team Stats

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2. Update the Pivot to show data for the Red Sox by year. Which years did they win the Division (DivWin)? The Wild Card (WCWin)? The World Series (WSWin)?

DivWin	Y						
WCWin	(All)						
WSWin	(All)						
Row Labels	RS	RA	Net Runs	HR	G	HR per Game	
Boston Red Sox	2,511	2,011	500	519	468	1.108974359	
1995	791	698	93	175	144	1.215277778	
2007	867	657	210	166	162	1.024691358	
2013	853	656	197	178	162	1.098765432	
Grand Total	2,511	2,011	500	519	468	1.108974359	

DivWin	(All)						
WCWin	Y						
WSWin	(All)						
Row Labels	RS	RA	Net Runs	HR	G	HR per Game	
Boston Red Sox	6,249	5,259	990	1425	1134	1.256613757	
1998	876	729	147	205	162	1.265432099	
1999	836	718	118	176	162	1.086419753	
2003	961	809	152	238	162	1.469135802	
2004	949	768	181	222	162	1.37037037	
2005	910	805	105	199	162	1.228395062	
2008	845	694	151	173	162	1.067901235	
2009	872	736	136	212	162	1.308641975	
Grand Total	6,249	5,259	990	1425	1134	1.256613757	

DivWin	(All)						
WCWin	(All)						
WSWin	Y						
Row Labels	RS	RA	Net Runs	HR	G	HR per Game	
Boston Red Sox	2,669	2,081	588	566	486	1.164609053	
2004	949	768	181	222	162	1.37037037	
2007	867	657	210	166	162	1.024691358	
2013	853	656	197	178	162	1.098765432	
Grand Total	2,669	2,081	588	566	486	1.164609053	

PROJECT 5: Baseball Team Stats

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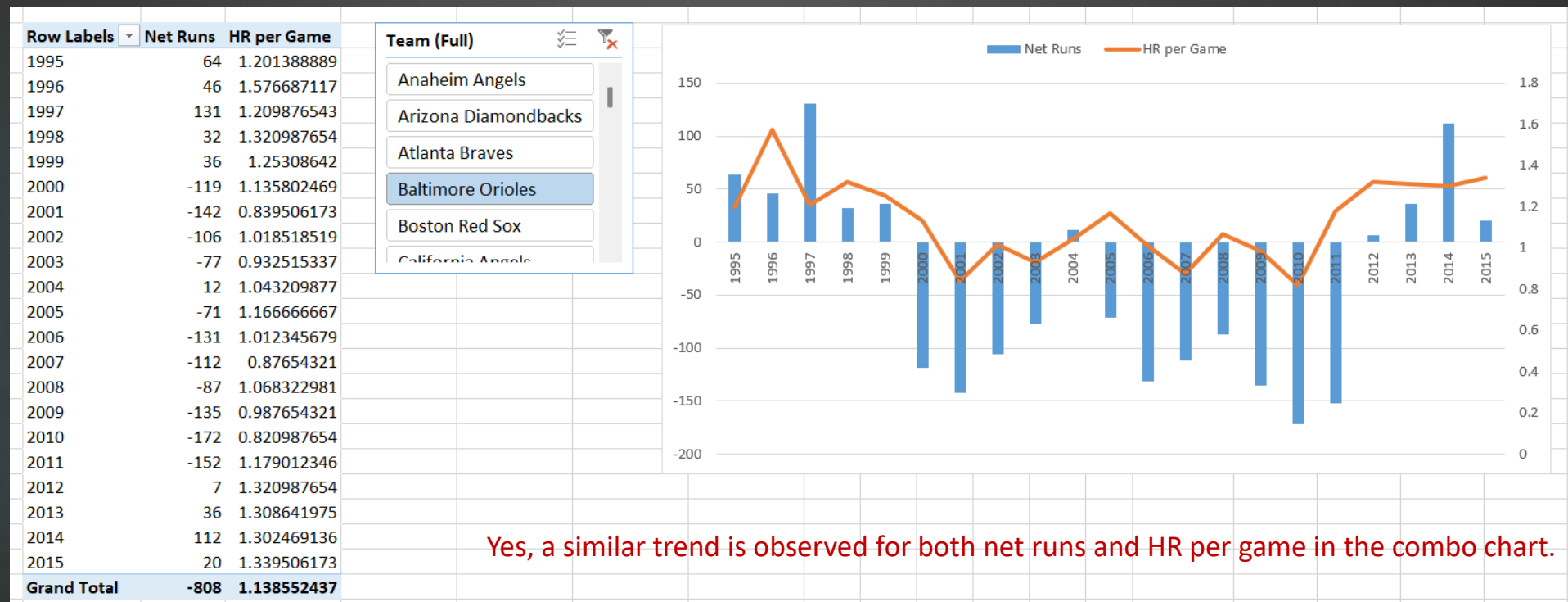
3. Show home runs (HR) by year, for the entire sample (all teams), and drag in a second instance as the % Difference From the previous year. In which season did overall home run totals decrease the most Y-o-Y?

Row Labels ▾	HR	% Difference From HR
1995	4,081	
1996	4,962	21.59%
1997	4,640	-6.49%
1998	5,064	9.14%
1999	5,528	9.16%
2000	5,693	2.98%
2001	5,458	-4.13%
2002	5,059	-7.31%
2003	5,207	2.93%
2004	5,451	4.69%
2005	5,017	-7.96%
2006	5,386	7.35%
2007	4,957	-7.97%
2008	4,878	-1.59%
2009	5,042	3.36%
2010	4,613	-8.51%
2011	4,552	-1.32%
2012	4,934	8.39%
2013	4,661	-5.53%
2014	4,186	-10.19%
2015	4,909	17.27%
Grand Total	104,278	

PROJECT 5: Baseball Team Stats

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4. Update your layout to show Net Runs and HR per Game as Values, Year as Row Labels, and Team (Full) as a Slicer. Select the Baltimore Orioles from the slicer, then insert a Pivot Chart formatted as a Combo Chart, with Net Runs plotted as columns on the primary axis, and HR per Game plotted as a line on the secondary axis. Do the two metrics follow a similar trend?



PROJECT 6: San Diego Burrito Ratings

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CASE STUDY: (Burrito ratings and Yelp reviews from 65 San Diego restaurants in 2016)

- ▶ Concepts: Calculating with COUNT; Summarize values by (SUM, AVERAGE); Show values as (Rank); Conditional Formatting
- ▶ Dimensions: Locations, Burrito, Date, recommendation, reviewer, Notes
- ▶ Measures: Yelp rating, Cost, Tortilla (0 – 5), Temp (0 – 5), Fillings (0 – 5), Meat Volume (0 - 5), Uniformity (0 – 5), Salsa (0 – 5), Synergy (0 – 5), Wrap Quality (0 – 5)

1. In your new PivotTable, compare average ratings for Tortilla, Temp, Fillings, Synergy, and Wrap Quality, by Location
2. Drag in the sum of the "# of Reviews" field and apply a value filter to only show locations with more than 2 ratings. How many locations recorded >2 ratings?
3. Create a calculated field named "Average Total Score", which correctly averages the five scores by location
4. Add a second instance of Average Total Score and show the values as a Rank (large to small) based on location. Among those with >2 reviews, which location is ranked #7?
5. Add a Colour Scale to the Average Total Score field and sort descending. Which location has the lowest score? The highest?
6. Drag in Yelp Rating as an average, add a Colour Scale, and compare against the Average Total Score field. How closely do the two fields align?

PROJECT 6: San Diego Burrito Ratings

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1. In your new PivotTable, compare average ratings for Tortilla, Temp, Fillings, Synergy, and Wrap Quality, by Location

Row Labels	Average of Tortilla (0-5)	Average of Temp (0-5)	Average of Fillings (0-5)	Average of Synergy (0-5)	Average of Wrap Quality (0-5)
Albertacos	3.5	3	3.75	3.75	4.5
Alberto's 623 N Escondido Blvd, Escondido, CA 92025	4	4	5	5	4
Burros and Fries	3.25	4.75	4	3.75	4.75
California Burritos	4.038095238	3.585714286	4	4.142857143	4.404761905
Cancun Mexican & Seafood	3.75	4.375	3.625	3.5	3.375
Carmen's Mexican Food	3.5	4.25	4	3.75	4.5
Chili Peppers	4	5	5	5	5
Chipotle	2.75	2.75	2.9	3.5	3.75
Colima's	5	5	5	5	4.5
Colima's Mexican Food	3.166666667	4.833333333	3.833333333	3.666666667	4
Donato's taco shop	3	5	3.5	4	4
El Cuervo	3.5	4.25	3.75	3.75	3.5
El dorado Mexican food	3.75	3.75	4	4	3.25
El Indio	4	4	4	3.5	4.5
El Nopalito	3.25	3.5	2.2	2.75	3.25
El Pueblo Mexican Food	4.5	4.5	4	4.5	5
El Rey Moro	3.5	4.5	3.5	3.5	5
El Torrito Foods	3.25	4.5	2.75	2.75	5
El Zarape	3.5	4.6	3.55	3.48	4.7
Goody's	2.666666667	2.333333333	2	1.833333333	1
Graciela's Taco Shop	3.75	4	3.25	4	3
Humbertos	4	5	5	5	5
Jorge's Mexicatessen	2.25	3.25	3.5	4	3.75
Juanita's Taco Shop	4.25	4.25	3.5	3.5	5
JV's Mexican Food	2.5	2	2.75	2.5	4
Karina's Taco Shop	3.75	4	3.5	3.75	3.5
King Burrito	2.75	3.75	3	2.5	4.25
Kotija Jr.	3	3	4	5	4
La Perla Cocina	4	3.833333333	3.666666667	2.833333333	4

PROJECT 6: San Diego Burrito Ratings

29

- Drag in the sum of the "# of Reviews" field and apply a value filter to only show locations with more than 2 ratings. How many locations recorded >2 ratings?

Row Labels	▼	Average of Tortilla (0-5)	Average of Temp (0-5)	Average of Fillings (0-5)	Average of Synergy (0-5)	Average of Wrap Quality (0-5)	Count of Reviewer
California Burritos		4.038095238	3.585714286	4	4.142857143	4.404761905	21
Cancun Mexican & Seafood		3.75	4.375	3.625	3.5	3.375	4
Colima's Mexican Food		3.166666667	4.833333333	3.833333333	3.666666667	4	3
El Zarape		3.5	4.6	3.55	3.48	4.7	5
Goody's		2.666666667	2.333333333	2	1.833333333	1	3
La Perla Cocina		4	3.833333333	3.666666667	2.833333333	4	3
Lolita's taco shop		2.983333333	3.275	3.641666667	3.4375	3.916666667	12
Los Primos Mexican Food		3.333333333	3.458333333	3.25	2.666666667	3.416666667	12
Los tacos		3.875	4.25	4	4.125	4.125	4
Lucha Libre North Park		3.678571429	3.2	3.364285714	3.314285714	4.014285714	28
Oscar's Mexican food		2.666666667	2.5	2.833333333	3.166666667	5	3
Porkyland		3.625	4.25	3.75	3.125	5	4
Rigoberto's Taco Shop		3.769230769	3.923076923	3.807692308	4	3.730769231	13
Roberto's Taco Shop Clairemont		3.7	3.8	3.25	3	4.875	4
Taco Stand		3.727272727	3.409090909	3.977272727	4.227272727	4.136363636	22
Taco Surf PB		4.166666667	2	4.133333333	4.066666667	4.666666667	3
Tacos por favor		3.333333333	4.833333333	3.166666667	3.666666667	2.333333333	3
Tony's Fresh Mexican Food		3.25	2.875	2.875	3.375	3.5	4
Vallarta express		2.916666667	4.055555556	3.444444444	3	3.611111111	9
Grand Total	14 Locations.	3.5696875	3.565	3.5971875	3.5540625	3.968125	160

PROJECT 6: San Diego Burrito Ratings

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3. Create a calculated field named "Average Total Score", which correctly averages the five scores by location

Row Labels	Average Total Score	Average of Tortilla (0-5)	Average of Temp (0-5)	Average of Fillings (0-5)	Average of Synergy (0-5)	Average of Wrap Quality (0-5)	Count of Reviewer
California Burritos	4.034285714	4.038095238	3.585714286	4	4.142857143	4.404761905	21
Cancun Mexican & Seafood	3.725	3.75	4.375	3.625	3.5	3.375	4
Colima's Mexican Food	3.9	3.166666667	4.833333333	3.833333333	3.666666667	4	3
El Zarape	3.966	3.5	4.6	3.55	3.48	4.7	5
Goody's	1.966666667	2.666666667	2.333333333	2	1.833333333	1	3
La Perla Cocina	3.666666667	4	3.833333333	3.666666667	2.833333333	4	3
Lolita's taco shop	3.450833333	2.983333333	3.275	3.641666667	3.4375	3.916666667	12
Los Primos Mexican Food	3.225	3.333333333	3.458333333	3.25	2.666666667	3.416666667	12
Los tacos	4.075	3.875	4.25	4	4.125	4.125	4
Lucha Libre North Park	3.514285714	3.678571429	3.2	3.364285714	3.314285714	4.014285714	28
Oscar's Mexican food	3.233333333	2.666666667	2.5	2.833333333	3.166666667	5	3
Porkyland	3.95	3.625	4.25	3.75	3.125	5	4
Rigoberto's Taco Shop	3.846153846	3.769230769	3.923076923	3.807692308	4	3.730769231	13
Roberto's Taco Shop Clairemont	3.725	3.7	3.8	3.25	3	4.875	4
Taco Stand	3.895454545	3.727272727	3.409090909	3.977272727	4.227272727	4.136363636	22
Taco Surf PB	3.806666667	4.166666667	2	4.133333333	4.066666667	4.666666667	3
Tacos por favor	3.466666667	3.333333333	4.833333333	3.166666667	3.666666667	2.333333333	3
Tony's Fresh Mexican Food	3.175	3.25	2.875	2.875	3.375	3.5	4
Vallarta express	3.405555556	2.916666667	4.055555556	3.444444444	3	3.611111111	9
Grand Total	3.6508125	3.5696875	3.565	3.5971875	3.5540625	3.968125	160

PROJECT 6: San Diego Burrito Ratings

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4. Add a second instance of Average Total Score and show the values as a Rank (large to small) based on location. Among those with >2 reviews, which location is ranked #7?

Row Labels	Average Total Score	Average of Total Score	Average of Tortilla (0-5)	Average of Temp (0-5)	Average of Fillings (0-5)	Average of Synergy (0-5)	Average of Wrap Quality (0-5)	Count of Reviewer
California Burritos	4.034285714	2	4.038095238	3.585714286	4	4.142857143	4.404761905	21
Cancun Mexican & Seafood	3.725	9	3.75	4.375	3.625	3.5	3.375	4
Colima's Mexican Food	3.9	5	3.166666667	4.833333333	3.833333333	3.666666667	4	3
El Zarape	3.966	3	3.5	4.6	3.55	3.48	4.7	5
Goody's	1.966666667	19	2.666666667	2.333333333	2	1.833333333	1	3
La Perla Cocina	3.666666667	11	4	3.833333333	3.666666667	2.833333333	4	3
Lolita's taco shop	3.450833333	14	2.983333333	3.275	3.641666667	3.4375	3.916666667	12
Los Primos Mexican Food	3.225	17	3.333333333	3.458333333	3.25	2.666666667	3.416666667	12
Los tacos	4.075	1	3.875	4.25	4	4.125	4.125	4
Lucha Libre North Park	3.514285714	12	3.678571429	3.2	3.364285714	3.314285714	4.014285714	28
Oscar's Mexican food	3.233333333	16	2.666666667	2.5	2.833333333	3.166666667	5	3
Porkyland	3.95	4	3.625	4.25	3.75	3.125	5	4
Rigoberto's Taco Shop	3.846153846	7	3.769230769	3.923076923	3.807692308	4	3.730769231	13
Roberto's Taco Shop Clairemont	3.725	10	3.7	3.8	3.25	3	4.875	4
Taco Stand	3.895454545	6	3.727272727	3.409090909	3.977272727	4.227272727	4.136363636	22
Taco Surf PB	3.806666667	8	4.166666667	2	4.133333333	4.066666667	4.666666667	3
Tacos por favor	3.466666667	13	3.333333333	4.833333333	3.166666667	3.666666667	2.333333333	3
Tony's Fresh Mexican Food	3.175	18	3.25	2.875	2.875	3.375	3.5	4
Vallarta express	3.405555556	15	2.916666667	4.055555556	3.444444444	3	3.611111111	9
Grand Total	3.6508125		3.5696875	3.565	3.5971875	3.5540625	3.968125	160

PROJECT 6: San Diego Burrito Ratings

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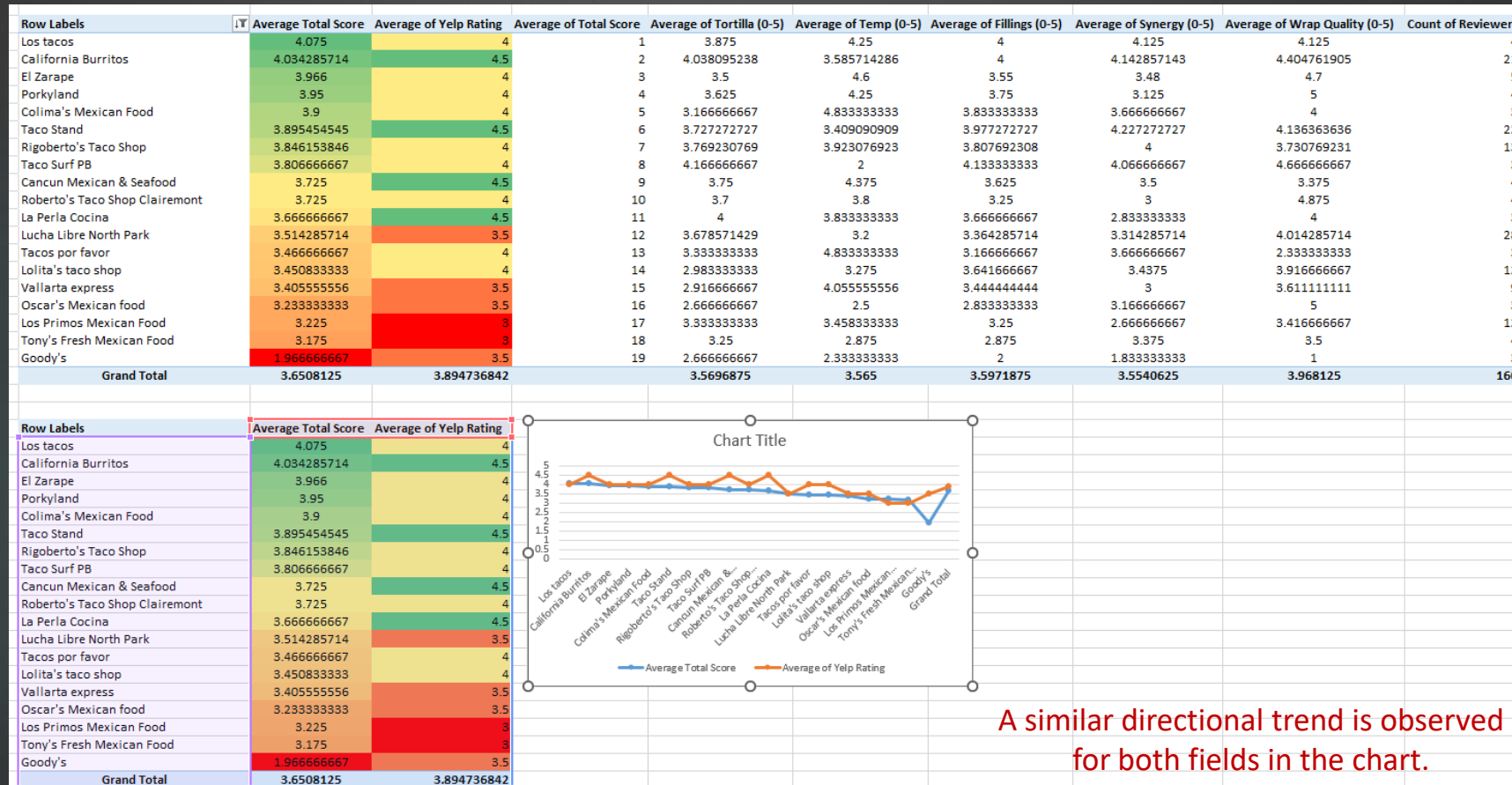
5. Add a Colour Scale to the Average Total Score field and sort descending. Which location has the lowest score? The highest?

Row Labels	Average Total Score	Average of Total Score	Average of Tortilla (0-5)	Average of Temp (0-5)	Average of Fillings (0-5)	Average of Synergy (0-5)	Average of Wrap Quality (0-5)	Count of Reviewer
Los tacos	4.075	1	3.875	4.25	4	4.125	4.125	4
California Burritos	4.034285714	2	4.038095238	3.585714286	4	4.142857143	4.404761905	21
El Zarape	3.966	3	3.5	4.6	3.55	3.48	4.7	5
Porkyland	3.95	4	3.625	4.25	3.75	3.125	5	4
Colima's Mexican Food	3.9	5	3.166666667	4.833333333	3.833333333	3.666666667	4	3
Taco Stand	3.895454545	6	3.727272727	3.409090909	3.977272727	4.227272727	4.136363636	22
Rigoberto's Taco Shop	3.846153846	7	3.769230769	3.923076923	3.807692308	4	3.730769231	13
Taco Surf PB	3.806666667	8	4.166666667	2	4.133333333	4.066666667	4.666666667	3
Cancun Mexican & Seafood	3.725	9	3.75	4.375	3.625	3.5	3.375	4
Roberto's Taco Shop Clairemont	3.725	10	3.7	3.8	3.25	3	4.875	4
La Perla Cocina	3.666666667	11	4	3.833333333	3.666666667	2.833333333	4	3
Lucha Libre North Park	3.514285714	12	3.678571429	3.2	3.364285714	3.314285714	4.014285714	28
Tacos por favor	3.466666667	13	3.333333333	4.833333333	3.166666667	3.666666667	2.333333333	3
Lolita's taco shop	3.450833333	14	2.983333333	3.275	3.641666667	3.4375	3.916666667	12
Vallarta express	3.405555556	15	2.916666667	4.055555556	3.444444444	3	3.611111111	9
Oscar's Mexican food	3.233333333	16	2.666666667	2.5	2.833333333	3.166666667	5	3
Los Primos Mexican Food	3.225	17	3.333333333	3.458333333	3.25	2.666666667	3.416666667	12
Tony's Fresh Mexican Food	3.175	18	3.25	2.875	2.875	3.375	3.5	4
Goody's	1.966666667	19	2.666666667	2.333333333	2	1.833333333	1	3
Grand Total	3.6508125		3.5696875	3.565	3.5971875	3.5540625	3.968125	160

PROJECT 6: San Diego Burrito Ratings

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6. Drag in Yelp Rating as an average, add a Color Scale, and compare against the Average Total Score field. How closely do the two fields align?



PROJECT 7: Daily Weather Conditions

34

CASE STUDY: (Daily Weather Conditions in Boston, Massachusetts from Jan – Dec 2016)

- ▶ Concepts: Sorting and Filtering; Pivotcharts; Date grouping; calculating with COUNT
- ▶ Dimensions: Date, Conditions
- ▶ Measures: Max Temp (F), Mean Temp (F), Min Temp (F), Max Wind Speed (MPH), Precipitation (in)

1. How many days in 2016 were categorized as Clear vs. Rain vs. Snow? (use the Conditions field)
2. What was the average temperature on clear days vs. snowy days? What about the average max temperature?
3. Update your view to show the # of Days by Month (primary row labels) and Conditions (secondary row labels), and show the # of Days as the % of the month as a whole. What percent of September days are clear?
4. Remove the second instance of # of Days, move Conditions to the column labels, and update the Show Values As calculation to % of Row Total. How often did it snow in January 2016, as a percentage of the month?
5. Remove grand totals and visualize the data as a 100% Stacked Column chart. In how many months of 2016 did it not snow at all?

PROJECT 7: Daily Weather Conditions

35

1. How many days in 2016 were categorized as Clear vs. Rain vs. Snow? (use the Conditions field)

Conditions	Count of Date
Clear	248
Rain	99
Snow	16
Grand Total	363

PROJECT 7: Daily Weather Conditions

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2. What was the average temperature on clear days vs. snowy days? What about the average max temperature?

Conditions ▼	Count of Date	Average of Mean Temp (F)	Average of Max Temp (F)
Clear	248	53.58064516	62.01209677
Rain	99	56.13131313	63.42424242
Snow	16	29.6875	36.625
Grand Total	363	53.2231405	61.27823691

PROJECT 7: Daily Weather Conditions

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3. Update your view to show the # of Days by Month (primary row labels) and Conditions (secondary row labels), and show the # of Days as the % of the month as a whole. What percent of September days are clear?

Count of Date		Conditions ▾			
Months (Date) ▾	Days (Date) ▾	Clear	Rain	Snow	Grand Total
⊕ Jan		77.42%	9.68%	12.90%	100.00%
⊕ Feb		65.52%	20.69%	13.79%	100.00%
⊕ Mar		58.06%	35.48%	6.45%	100.00%
⊕ Apr		66.67%	26.67%	6.67%	100.00%
⊕ May		67.74%	32.26%	0.00%	100.00%
⊕ Jun		76.67%	23.33%	0.00%	100.00%
⊕ Jul		74.19%	25.81%	0.00%	100.00%
⊕ Aug		74.19%	25.81%	0.00%	100.00%
⊕ Sept		63.33%	36.67%	0.00%	100.00%
⊕ Oct		64.52%	35.48%	0.00%	100.00%
⊕ Nov		66.67%	33.33%	0.00%	100.00%
⊕ Dec		64.29%	21.43%	14.29%	100.00%
Grand Total		68.32%	27.27%	4.41%	100.00%

PROJECT 7: Daily Weather Conditions

38

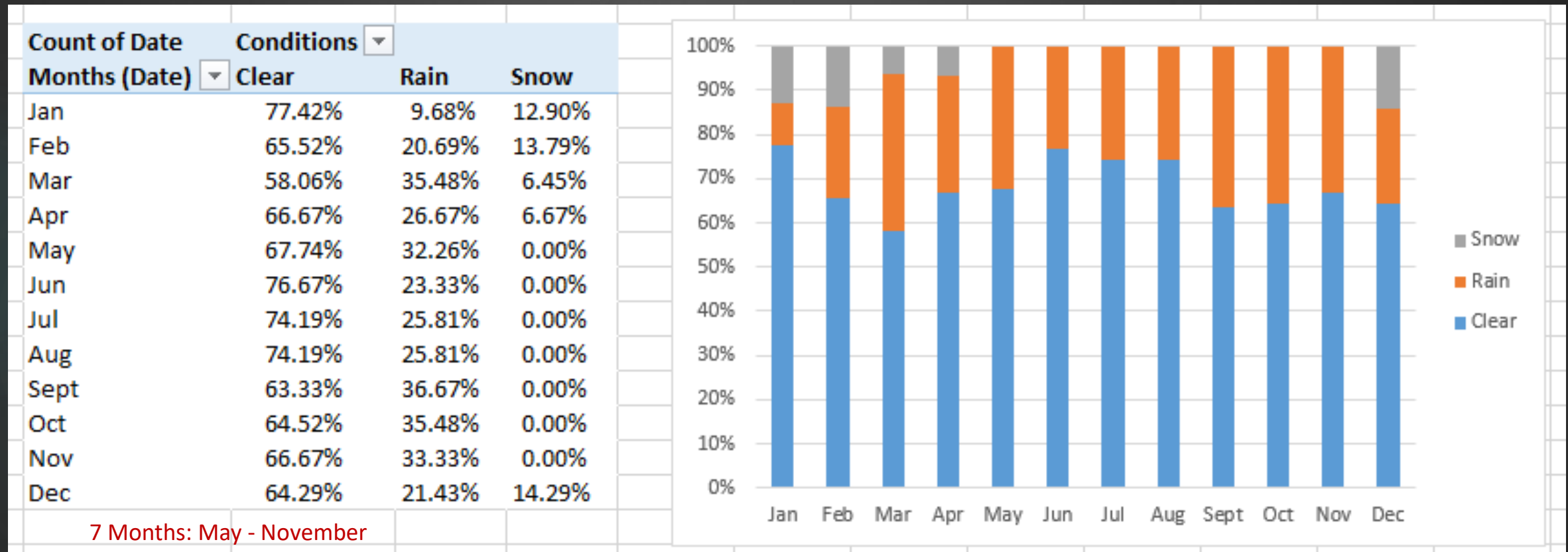
4. Remove the second instance of # of Days, move Conditions to the column labels, and update the Show Values As calculation to % of Row Total. How often did it snow in January 2016, as a percentage of the month?

Count of Date	Conditions			
Months (Date)	Clear	Rain	Snow	Grand Total
Jan	77.42%	9.68%	12.90%	100.00%
Feb	65.52%	20.69%	13.79%	100.00%
Mar	58.06%	35.48%	6.45%	100.00%
Apr	66.67%	26.67%	6.67%	100.00%
May	67.74%	32.26%	0.00%	100.00%
Jun	76.67%	23.33%	0.00%	100.00%
Jul	74.19%	25.81%	0.00%	100.00%
Aug	74.19%	25.81%	0.00%	100.00%
Sept	63.33%	36.67%	0.00%	100.00%
Oct	64.52%	35.48%	0.00%	100.00%
Nov	66.67%	33.33%	0.00%	100.00%
Dec	64.29%	21.43%	14.29%	100.00%
Grand Total	68.32%	27.27%	4.41%	100.00%

PROJECT 7: Daily Weather Conditions

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5. Remove grand totals and visualize the data as a 100% Stacked Column chart. In how many months of 2016 did it not snow at all?



PROJECT 8: Spartan Race Facebook Posts

40

CASE STUDY: (Spartan Race Facebook posts from Aug – Oct 2016)

- ▶ Concepts: Show Value as (% of Column); calculated Fields; Calculating with COUNT
- ▶ Dimensions: page, Date of post, post copy, link, post type, hour of post, time of day
- ▶ Measures: Shares, reactions, likes, Comments

1. Create a new view showing the # of Posts by Date of Post. On which date did Spartan Race post most often? What types of posts were they?
2. Which date showed the highest total engagement volume? Highest engagements per post? On the date with the highest engagements per post, which one drove the strongest response?
3. Create a new calculated field named "Sharability" ($\text{Shares} / \text{Total Engagements}$). Which post types tend to be most "sharable"? What time of day do people tend to share most?
4. Create a new calculated field named "Active Engagements" ($\text{Shares} + \text{Comments}$), then modify the formula to measure Active Engagements per Post. What time of day are users LEAST likely to actively engage with Spartan Race Facebook posts? How many active engagements per post during that time of day?

PROJECT 8: Spartan Race Facebook Posts

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1. Create a new view showing the # of Posts by Date of Post. On which date did Spartan Race post most often? What types of posts were they?

Date of Post	Post Type	Count of Post Copy
01/10/2016		25
	photo	4
	video	21
30/09/2016		17
	photo	5
	video	12
02/10/2016		17
	event	1
	link	1
	photo	7
	video	8

PROJECT 8: Spartan Race Facebook Posts

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2. Which date showed the highest total engagement volume? Highest engagements per post? On the date with the highest engagements per post, which one drove the strongest response?

Date of Post	Post Copy	Sum of Total Engagements	Sum of Engagements per Post
11/09/2016		16,700	3,340
	We remember.	13,224	13,224
		0	1,596
	Mountains, mud and... more mud. #AROO to the Spartans who braved the #Trifecta weekend in Tirol, Aus...	1,332	1,332
	Congratulations to the top Elites of the 2016 #SpartanRace Ft Bragg Sprint.Your top three Elite wom...	479	479
	Get in gear for your next challenge! SAVE 25% off the Peak Mountain Bike Gnarly Adventure or 6 Hour ...	69	69
21/09/2016		19,759	3,293
	We're not 100% positive that he's performing a full push-up at the bottom of his burpee, but we'll g...	15,843	15,843
	When you know you look good just after completing a muddy obstacle. #NotAModel #ActualRacer #WCW #Sp...	2,207	2,207
	#SpartanParents- teaching kids to throw pointy objects since 431 BC.	826	826
	Tell your story! #SpartanRace #WHYIRACEhttp://sprtn.im/WHYIRACE_2016	335	335
	Coach Jodie Somers is live from Australia to take you through some mobility transitions to help keep...	310	310
	Coach Q "Always Learning" XPT Training	238	238
09/10/2016		9,675	2,419
	After finishing the race, they started their life together. Give an "AROO!" to this happy #SpartanCo...	6,034	6,034

PROJECT 8: Spartan Race Facebook Posts

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3. Create a new calculated field named "Shareability" (Shares / Total Engagements). Which post types tend to be most "sharable"? What time of day do people tend to share most?

Post Type ▾	Sum of Total Engagements	Sum of Engagements per Post	Sum of Shareability
video	143,065	734	9.58%
link	8,935	288	6.91%
photo	147,354	945	5.50%
status	227	114	1.76%
event	933	104	0.00%
Grand Total	300,514	765	7.46%

Time of Day ▾	Sum of Total Engagements	Sum of Engagements per Post	Sum of Shareability
Late Night	65,360	787	9.53%
Evening	155,601	724	7.06%
Afternoon	65,007	823	6.91%
Morning	14,546	909	4.90%
Grand Total	300,514	765	7.46%

PROJECT 8: Spartan Race Facebook Posts

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4. Create a new calculated field named "Active Engagements" (Shares + Comments), then modify the formula to measure Active Engagements per Post. What time of day are users LEAST likely to actively engage with Spartan Race Facebook posts? How many active engagements per post during that time of day?

Time of Day ▾	Sum of Active Engagements	Sum of Active Engagements per Post	Sum of Engagements per Post
Morning	1,256	78.50	909
Evening	17,565	81.70	724
Afternoon	7,418	93.90	823
Late Night	10,787	129.96	787
Grand Total	37,026	94.21	765

PROJECT 9: Apple App Data

45

CASE STUDY: (Details and descriptions for 7,000+ mobile apps available on the Apple App Store)

- ▶ Concepts: Sorting and Filtering, Calculated Fields, Show Value as (% of Column, % of Parent); Conditional Formatting
- ▶ Dimensions: App ID, App Name, Size, Currency, Price, Age Rating, Primary Genre
- ▶ Measures: Total ratings, Total average rating, Current Version Ratings, Current average Rating, Supported Devices, # of Screenshots, Supported Languages

1. Add a calculated column in the source data named "Volume", equal to "High" for rows where Total Ratings > 100,000 and "Low" otherwise
 - a. Clear the existing PivotTable and create a new view showing Volume as row labels, with the count of App Name and the sum of Total Ratings as values. How many apps are categorized as high-volume?
 - b. Update both fields to show values as % of Column Total. What percentage of all apps are high-volume, and what percentage of all ratings do those high-volume apps account for?
 - c. How do average ratings compare between high-volume and low-volume apps?
2. Pull in the App Name as secondary row labels, remove the Count of App Name field, and filter your view to show the Top 10 apps descending by Total Ratings.
 - a. Among high-volume apps, which 3 drove the most ratings?
 - b. Among apps where the primary genre is "Games", which apps drove more than 1 million ratings? (Hint: you may need to change your value calculation).
 - c. Add a second instance of the Sum of Total Ratings field, and use conditional formatting to visualize the values for high-volume game apps as data bars, with no overlapping text
3. Add a new calculated column in the source data named "Rating Improvement" (Current Version Avg Rating - Avg Rating).
 - a. Among high-volume Entertainment apps specifically, which app saw the largest rating improvement with the current version?
 - b. Did any of these apps see a decline with the current version? If so, what did the rating drop to?

PROJECT 9: Apple App Data

46

1. Add a calculated column in the source data named "Volume", equal to "High" for rows where Total Ratings > 100,000 and "Low" otherwise
 - a. Clear the existing PivotTable and create a new view showing Volume as row labels, with the count of App Name and the sum of Total Ratings as values. How many apps are categorized as high-volume?
 - b. Update both fields to show values as % of Column Total. What percentage of all apps are high-volume, and what percentage of all ratings do those high-volume apps account for?
 - c. How do average ratings compare between high-volume and low-volume apps?

Volume ▾	Count of App Name	Sum of Total Ratings
High	218	61,992,054
Low	6,979	30,798,199
Grand Total	7,197	92,790,253

Volume ▾	% of App Name Column Total	% of Total Ratings Column Total
High	3.03%	66.81%
Low	96.97%	33.19%
Grand Total	100.00%	100.00%

Volume ▾	Average of Avg Rating	% of Avg Rating Column Total	% of App Name Column Total
High	4.268348624	3.67%	3.03%
Low	3.503797106	96.33%	96.97%
Grand Total	3.526955676	100.00%	100.00%

PROJECT 9: Apple App Data

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2. Pull in the App Name as secondary row labels, remove the Count of App Name field, and filter your view to show the Top 10 apps descending by Total Ratings.
 - a. Among high-volume apps, which 3 drove the most ratings?
 - b. Among apps where the primary genre is "Games", which apps drove more than 1 million ratings? (Hint: you may need to change your value calculation).
 - c. Add a second instance of the Sum of Total Ratings field, and use conditional formatting to visualize the values for high-volume game apps as data bars, with no overlapping text

Volume	App Name	Sum of Total Ratings
High		14,830,816
	Facebook	2,974,676
	Instagram	2,161,558
	Clash of Clans	2,130,805
	Temple Run	1,724,546
	Pandora - Music & Radio	1,126,879
	Pinterest	1,061,624
	Bible	985,920
	Candy Crush Saga	961,794
	Spotify Music	878,563
	Angry Birds	824,451
Low		980,554
	The Calculator - Free and Easy Calculating!	99,244
	Shadow Fight 2	99,206
	Flick Home Run !	98,851
	Voice Changer Plus	98,777
	iFunny :)	98,344
	Terraria	98,036
	The CW	97,368
	War Robots	97,122
	MeetMe - Chat and Meet New People	97,072
	Lumosity - Brain Training	96,534
Grand Total		15,811,370

Primary Genre	Games			
Volume	App Name	Average of Avg Rating	Sum of Total Ratings	Sum of Total Ratings2
High		4.5	9,684,135	9,684,135
	Clash of Clans	4.5	2,130,805	
	Temple Run	4.5	1,724,546	
	Candy Crush Saga	4.5	961,794	
	Angry Birds	4.5	824,451	
	Subway Surfers	4.5	706,110	
	Fruit Ninja Classic	4.5	698,516	
	Solitaire	4.5	679,055	
	CSR Racing	4.5	677,247	
	Crossy Road - Endless Arcade Hopper	4.5	669,079	
	Injustice: Gods Among Us	4.5	612,532	
Low		4.3	931,051	931,051
	Shadow Fight 2	4.5	99,206	99,206
	Flick Home Run !	4.5	98,851	98,851
	Terraria	4.5	98,036	98,036
	War Robots	4.5	97,122	97,122
	Angry Birds Space	4	95,892	95,892
	PewDiePie's Tuber Simulator	5	90,851	90,851
	Angry Birds HD	4.5	89,110	89,110
	Super Mario Run	2.5	87,958	87,958
	Tap Titans	4.5	87,898	87,898
	Scribblenauts Remix	4.5	86,127	86,127

Primary Genre	Games	
Volume	App Name	Sum of Total Ratings
High		3,855,351
	Clash of Clans	2,130,805
	Temple Run	1,724,546

PROJECT 9: Apple App Data

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3. Add a new calculated column in the source data named "Rating Improvement" (Current Version Avg Rating - Avg Rating).
 - a. Among high-volume Entertainment apps specifically, which app saw the largest rating improvement with the current version?
 - b. Did any of these apps see a decline with the current version? If so, what did the rating drop to?

Primary Genre	Entertainment				
Volume	App Name	Sum of Rating Improvement	Sum of Current version Avg Rating	Average of Avg Rating	Sum of Total Ratings
High		3	38.5	3.944444444	1,719,109
	Mad Libs	1.5	4.5	3	117,889
	Fandango Movies - Times + Tickets	1	5	4	291,787
	TRUTH or DARE!!! - FREE	0.5	3.5	3	171,055
	IMDb Movies & TV - Trailers and Showtimes	0.5	5	4.5	183,425
	Colorfy: Coloring Book for Adults	0	4.5	4.5	247,809
	Twitch	0	4.5	4.5	109,549
	Pocket God	0	4	4	187,529
	Action Movie FX	0	4.5	4.5	101,222
	Netflix	-0.5	3	3.5	308,844
Low		3	38	3.5	787,990
	Redbox	1.5	4.5	3	60,236
	Voice Changer Plus	1	4	3	98,777
	iFunny :)	1	4.5	3.5	98,344
	The Moron Test	0.5	4.5	4	88,613
	ABC æ" Watch Live TV & Stream Full Episodes	0.5	3.5	3	78,890
	Hulu: Watch TV Shows & Stream the Latest Movies	0	2	2	56,170
	Xbox	0	4.5	4.5	72,187
	Talking Tom Cat 2 for iPad	0	4	4	56,399
	DIRECTV	-0.5	3	3.5	81,006
	The CW	-1	3.5	4.5	97,368

PROJECT 10: Wine Tasting Scores

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CASE STUDY: (Data from 130,000 wine tastings, including locations, wine description, points and price)

- ▶ Concepts: Sorting and Filtering, Value Grouping, Pivotcharts
- ▶ Dimensions: Country, Province, Region, Subregion, Winery, Variety, Wine Name, Taster Name, Taster Twitter Handle, Description
- ▶ Measures: Points, Price

1. Ungroup the price values, and create a new view showing the average point rating and average price with Subregion as row labels
 - a. Which subregion in the sample produced the highest-rated wines, on average?
 - b. Which subregion in the sample produced the highest-priced wines, on average?
 - c. Excluding blanks, which subregion has the most tastings recorded in the dataset?
2. Create a new view to show the average point ratings by Country (sorted descending by average points), and add a PivotChart to visualize the relative ratings as a Clustered Column Chart
 - a. On average, which country has generated the highest average ratings? The lowest?
 - b. How do the ratings compare across countries for Pinot Grigio varieties specifically?
3. Delete the PivotChart, then update your table to show the actual Wine Name for all Italian wines in the sample, along with the point rating and price
 - a. What's the cheapest Italian wine rated over 95 points?
 - b. Who reviewed that particular wine, and how did he/she describe it?

PROJECT 10: Wine Tasting Scores

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1. Ungroup the price values, and create a new view showing the average point rating and average price with Subregion as row labels
 - a. Which subregion in the sample produced the highest-rated wines, on average?
 - b. Which subregion in the sample produced the highest-priced wines, on average?
 - c. Excluding blanks, which subregion has the most tastings recorded in the dataset?

Row Labels	Average of Points	Average of Price
Willamette Valley	89.46	\$39.48
Napa	89.39	\$60.66
Columbia Valley	88.99	\$32.81
Sonoma	88.93	\$39.86
Central Coast	88.83	\$35.74
Napa-Sonoma	88.79	\$40.12
Oregon Other	88.58	\$30.68
(blank)	88.33	\$34.16
Washington Other	88.27	\$26.55
Sierra Foothills	88.27	\$27.01
North Coast	87.98	\$30.38
South Coast	87.83	\$42.83
Long Island	87.56	\$29.45
Southern Oregon	87.44	\$26.71
Finger Lakes	87.13	\$20.51
Central Valley	86.93	\$19.38
New York Other	86.44	\$21.17
California Other	85.70	\$16.17
Grand Total	88.45	\$35.36

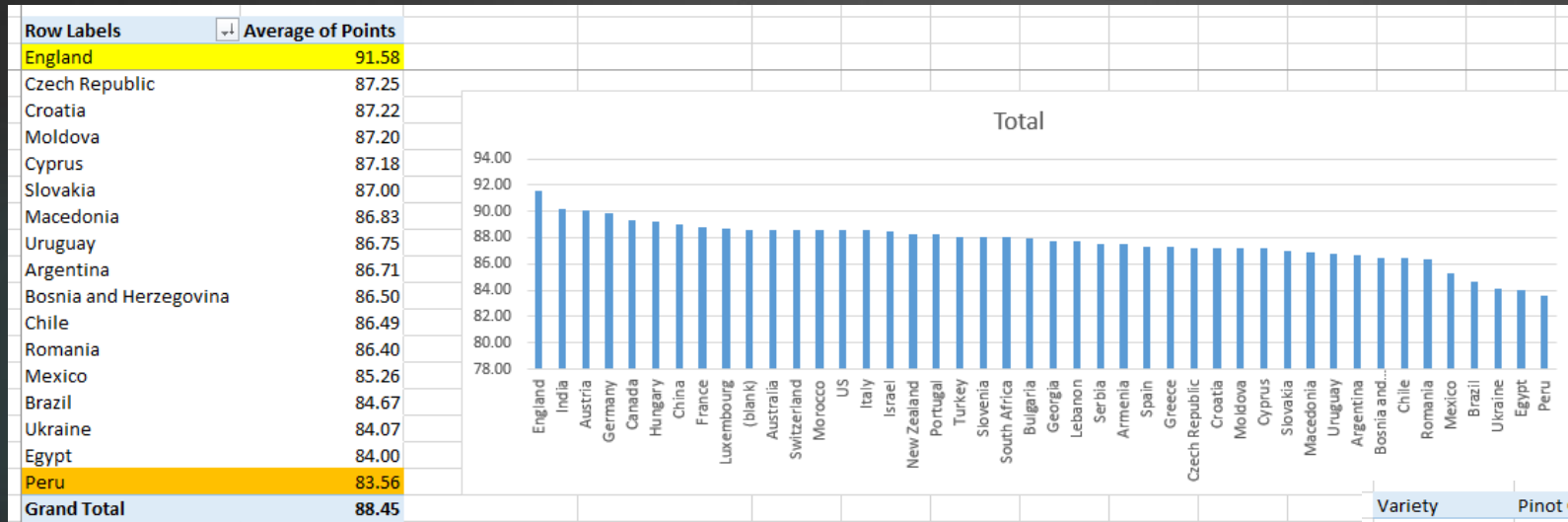
Row Labels	Average of Points	Average of Price
Napa	89.39	\$60.66
South Coast	87.83	\$42.83
Napa-Sonoma	88.79	\$40.12
Sonoma	88.93	\$39.86
Willamette Valley	89.46	\$39.48
Central Coast	88.83	\$35.74
(blank)	88.33	\$34.16
Columbia Valley	88.99	\$32.81
Oregon Other	88.58	\$30.68
North Coast	87.98	\$30.38
Long Island	87.56	\$29.45
Sierra Foothills	88.27	\$27.01
Southern Oregon	87.44	\$26.71
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Finger Lakes	87.13	\$20.51
Central Valley	86.93	\$19.38
California Other	85.70	\$16.17
Grand Total	88.45	\$35.36

Row Labels	Count of Tasting ID	Average of Points	Average of Price
Central Coast	11,065	88.83	\$35.74
Sonoma	9,028	88.93	\$39.86
Columbia Valley	8,103	88.99	\$32.81
Napa	6,814	89.39	\$60.66
Willamette Valley	3,423	89.46	\$39.48
California Other	2,663	85.70	\$16.17
Finger Lakes	1,777	87.13	\$20.51
Sierra Foothills	1,462	88.27	\$27.01
Napa-Sonoma	1,169	88.79	\$40.12
Central Valley	1,062	86.93	\$19.38
Southern Oregon	917	87.44	\$26.71
Oregon Other	727	88.58	\$30.68
Long Island	680	87.56	\$29.45
North Coast	584	87.98	\$30.38
Washington Other	534	88.27	\$26.55
South Coast	272	87.83	\$42.83
New York Other	231	86.44	\$21.17
Grand Total	50,511	88.63	\$37.06

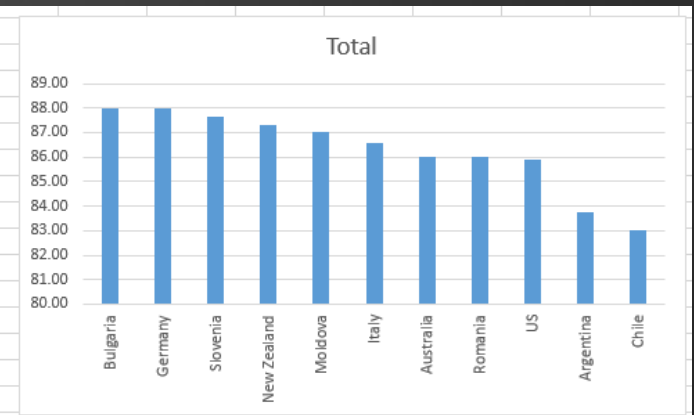
PROJECT 10: Wine Tasting Scores

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2. Create a new view to show the average point ratings by Country (sorted descending by average points), and add a PivotChart to visualize the relative ratings as a Clustered Column Chart
 - a. On average, which country has generated the highest average ratings? The lowest?
 - b. How do the ratings compare across countries for Pinot Grigio varieties specifically?



Variety	Pinot Grigio
Row Labels	Average of Points
Bulgaria	88.00
Germany	88.00
Slovenia	87.67
New Zealand	87.33
Moldova	87.00
Italy	86.56
Australia	86.00
Romania	86.00
US	85.89
Argentina	83.78
Chile	83.00
Grand Total	86.24



PROJECT 10: Wine Tasting Scores

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3. Delete the PivotChart, then update your table to show the actual Wine Name for all Italian wines in the sample, along with the point rating and price
 - a. What's the cheapest Italian wine rated over 95 points?
 - b. Who reviewed that particular wine, and how did he/she describe it?

Variety	(All)	
Country	Italy	
Wine Name	⌵ Average of Points	Average of Price
Cavallotto 2011 Bricco Boschis Vigna San Giuseppe Riserva (Barolo)	96.00	
Principe Corsini Fattoria Le Corti 1997 Sant'Andrea Corsini (Vin Santo del Chianti Classico)	97.00	
Isole e Olena 2010 Chianti Classico	96.00	27
Colombera & Garella 2013 Pizzaguerra (Lessona)	96.00	35

Variety	(All)			
Country	Italy			
Wine Name	⌵	Taster Name	Description	Average of Points
⊞ Cavallotto 2011 Bricco Boschis Vigna San Giuseppe Riserva (Barolo)		⊞ Kerin Oâ€™Keefe		96.00
			New leather, tilled earth, dark spice, underbrush and a balsan	96.00
⊞ Principe Corsini Fattoria Le Corti 1997 Sant'Andrea Corsini (Vin Santo del Chianti Classico)		⊞ (blank)		97.00
			A true labor of love, this fabulous Vin Santo opens with beaut	97.00
⊞ Isole e Olena 2010 Chianti Classico		⊞ Kerin Oâ€™Keefe		96.00
			This stunning Chianti Classico has an intense perfume of violet, rose and iris, accented with scents of truffle and forest berry. The wildly delicious palate presents a core of juicy cherry layered with sweet pipe tobacco, mint and spice,	96.00
⊞ Colombera & Garella 2013 Pizzaguerra (Lessona)		⊞ Kerin Oâ€™Keefe		96.00
				35



PROJECT: Data Analysis with Excel Pivot Tables

PROJECT - END